

Kunal Mathur

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GitHub: github.com/kunal202426 | Portfolio: portfolio-kunal26.vercel.app

SUMMARY

AI-native full-stack engineer: builds AI-powered features at Astra using the Vercel AI SDK with Google Gemini and DeepSeek (structured output via Zod, streaming responses), including AI resume/GitHub/portfolio analysers and an AI "explain this page" assistant. Builds agent infrastructure on the side: a GraphRAG memory layer exposing a 9-tool agentic API over MCP (CodeGraph), a multi-agent coordination system (Autonomous Agent Marketplace), and a production multi-provider LLM router with automatic fallback (JobHuntPro). Also trains models directly: a real-time computer-vision pipeline (MobileNetV2 CNN, ResNet-18, XGBoost) and a PyTorch fault-detection system. IIT Madras Global Hyperloop DesignX winner; Flipkart GRiD 7.0 national semifinalist.

EXPERIENCE

Astra

Founding Engineer

Sep 2026 – Present

Bengaluru, India (On-site)

- Builds AI-powered product features (Vercel AI SDK, Google Gemini, DeepSeek, structured output with Zod, streaming responses) across resume/GitHub/portfolio analysers and an AI page-explainer assistant, on Next.js 16 / React 19 / Supabase.
- Building a new recruiter-facing area end to end: UI, a typed service layer, and a multi-tenant data model.

YES Securities (YES Bank)

Full Stack Developer Intern

Jan 2026 – Aug 2026

Bengaluru, India

- Shipped 5+ production features (React, Node.js, REST APIs) on the OMNI trading platform serving 20,000+ daily trades.

PROJECTS

CodeGraph: GraphRAG Memory Layer for AI Coding Agents | GitHub

2026 – Present

- Local-first memory and retrieval layer for AI agents: semantic embeddings fused with literal search via Reciprocal Rank Fusion over a DuckDB entity/edge graph, exposed as a 9-tool agentic API over MCP; cuts agent context usage 9.6x.

Eye Tracking Capstone: Real-Time CV Pipeline | GitHub

2026

- Real-time webcam pipeline (~30fps on GPU) classifying emotion, attention (MobileNetV2 CNN), and cognitive load (XGBoost), plus gaze estimation (ResNet-18); CNN/gaze models exported to ONNX for a measured 4-30x inference speedup.

JobHuntPro: Browser Extension + Web Platform | Live

2026 – Present

- Production LLM system with multi-provider routing (Claude, Gemini, OpenRouter) and automatic fallback, live and in daily use.

EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science & Engineering

2022 – 2026

Vellore, Tamil Nadu, India

ACHIEVEMENTS

Winner: Global Hyperloop Competition (Asia's First), IIT Madras

2025

- AI & Software Lead, Team Levitate (VIT): led the software workstream; team won the DesignX trophy and 5 category awards against 15+ international teams.

National Semifinalist: Flipkart GRiD 7.0

2025

SKILLS

AI & Agents: LLM APIs (Claude, Gemini, DeepSeek), Vercel AI SDK, MCP, Multi-Agent Orchestration, Prompt Engineering, Zod

ML & Computer Vision: PyTorch, CNNs (MobileNetV2), ResNet-18, XGBoost, ONNX, LSTM, LightGBM, FinBERT

Backend & Data: Node.js, FastAPI, Supabase, PostgreSQL, MongoDB, REST APIs

Frontend & Tools: Next.js, React, TypeScript, Git, Agile